Heterogeneous Knowledge Distillation for Real-Time 2D Hand Pose Estimation

From HRNetV2-W18 Heatmaps to BlazeHandLandmark Coordinate Regression

Bea Juliana B. Poquiz · Rhandley D. Cajote, Ph.D. · Dale Joshua R. Del Carmen

THE PROBLEM

MediaPipe Hands
Real-time ✓ · Widely deployed ✓
Non-retrainable ✗ · Closed-source ✗

Researchers cannot:

- Fine-tune for surgical tracking
- Adapt to sign-language domains
- Inspect training data or losses
- Evaluate on public benchmarks

"You cannot retrain Google's MediaPipe."

OBJECTIVES

① **Build an open KD pipeline**
Fully retrainable · No proprietary data needed · Complete training code released

② **Resolve representation mismatch**
Bridge heatmap teacher → coordinate student via MSRAHeatmap codec + Taylor expansion

③ **Produce a MediaPipe alternative**
100% detection · PCK@0.2 = 0.748 · 139 FPS · retargetable to any domain

DATASET: RHD2D

Rendered Handpose Dataset (RHD2D)
Fully synthetic · pixel-perfect GT annotations · no manual labelling

Property	Value
Training images	41,258
Test images	2,728
Input resolution	256 × 256 px
Keypoints per image	21
Annotation types	2D/3D coords, visibility, depth

21 KEYPOINTS TOPOLOGY
Wrist (×1) + MCP · PIP · DIP · Tip per finger (×5 fingers)
Index · Middle · Ring · Pinky · Thumb → 21 joints total

METHODOLOGY: HETEROGENEOUS KNOWLEDGE DISTILLATION PIPELINE

RHD2D Dataset (41,258 train, 256×256 px) → **TEACHER HRNetV2-W18** (9.64M params, 21×64×64 heatmaps) → **MSRAHeatmap CODEC (DARK)** (Taylor expansion, Sub-pixel decode) → **STUDENT BlazeHandLandmark** (2.01M params, 21×2 coordinates) → **OUTPUT Checkpoint** (139 FPS · 7.75 MB, distilled_v2_ft.pth)

Wing Loss + Bone Loss (λ=0.002)

PHASE 1 — DISTILLATION (Epochs 1–100)
lr = 1×10^{-4} · Adam · Batch 32 · Cosine annealing · Augmentation: flip, bbox jitter, photometric distortion
Init: MediaPipe pretrained weights · Loss: Wing + 0.002×Bone vs. teacher-decoded coords

PHASE 2 — FINE-TUNING (Epochs 101–150)
lr = 5×10^{-5} · Adam · Gradient clipping (norm=5.0) · Early stopping (patience=25 epochs)
Output: distilled_v2_ft_epoch_150.pth (7.75 MB) · No ground-truth labels used anywhere

KEY CONTRIBUTION: heterogeneous KD framework — heatmap teacher (HRNetV2-W18) → coordinate student (BlazeHandLandmark) bridged via MSRAHeatmap Codec with second-order Taylor refinement

0.748	0.723	36.87 px	100%
PCK@0.2	AUC at ep.100	MPJPE at ep.100	Final det. rate

QUANTITATIVE RESULTS

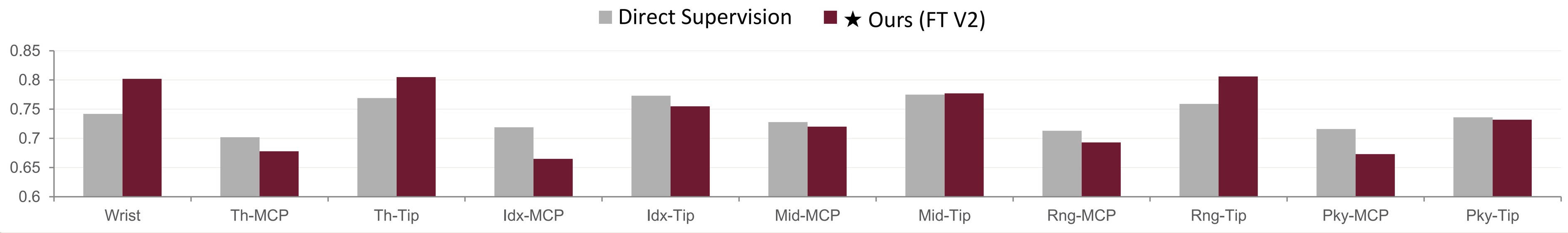
Model	Params	PCK@0.2 ↑	AUC ↑	MPJPE↓(px)	Det.Rate
MediaPipe SDK	1.98M	0.223	0.307	133.27	66.23%
BlazeHand (pre-KD)	2.01M	0.301	0.395	104.29	80.05%
Direct Supervision	2.01M	0.745	0.686	40.34	99.96%
FineTuned V2	**2.01M**	**0.748**	**0.723**	**36.87**	**100%**
Teacher HRNetV2†	9.64M	0.992	0.902	2.21	100%

† Teacher is upper-bound reference only; 6.9 FPS disqualifies deployment. All metrics on full RHD2D test set (2,727 images). Detection failures scored at 256 px/joint.

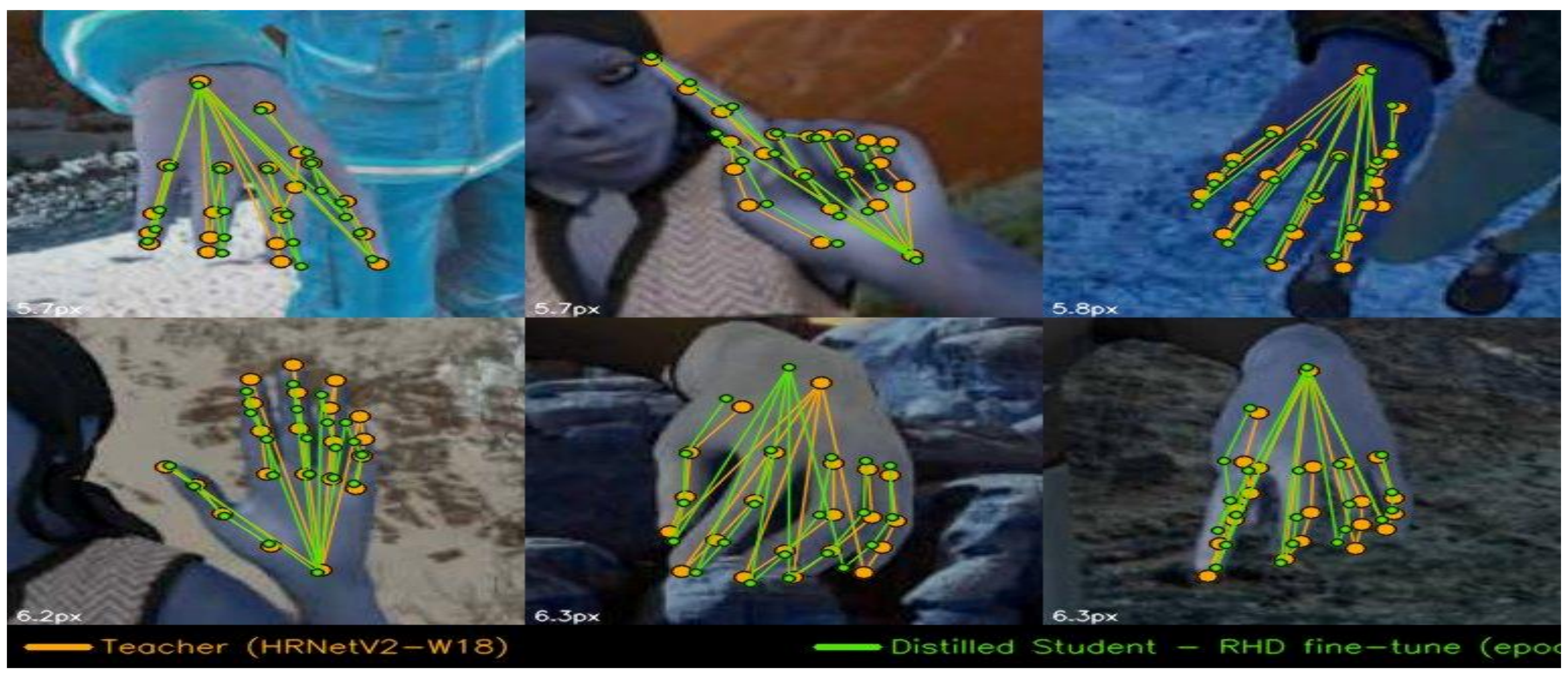
Distillation beats direct supervision: +3.7 pp AUC · −3.47 px MPJPE · same 139 FPS

139 FPS	4.8×	20.2×
GPU Inference (RTX 4050)	Parameter Compression	Full-Pipeline Speedup

Per-Joint PCK@0.2: Distilled Student vs. Direct Supervision



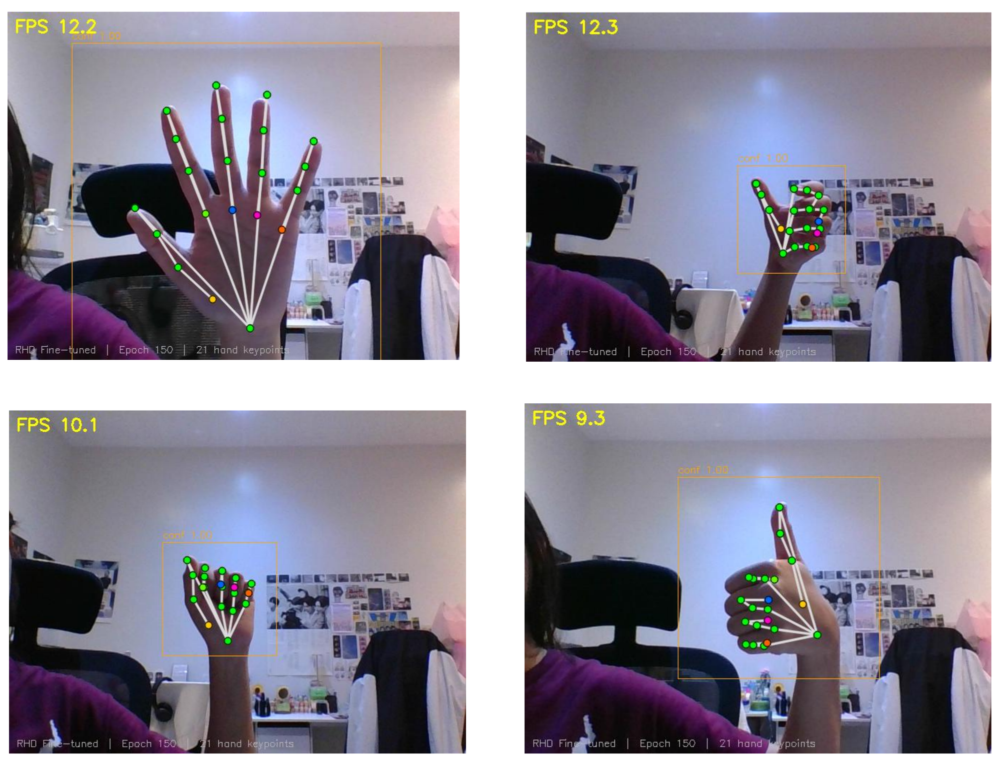
COMPUTER VISION DEMO — MODEL IN ACTION



● Teacher (HRNetV2-W18) ● Distilled Student (FT V2 ep.150)
21 anatomical keypoints predicted per frame · Varied poses, partial occlusion, synthetic + real-world hands

5.7 px EPE	5.7 px EPE	5.8 px EPE	6.2 px EPE	6.3 px EPE	6.3 px EPE
Top-left	Top-center	Top-right	Bottom-left	Bottom-center	Bottom-right

Out-of-distribution generalization: A model trained on flat-shaded synthetic RHD2D data generalizes directly to real webcam inputs without domain-specific fine-tuning (see above frame and webcam images below of the researcher's hand).



SCAN FOR DEMO

OPEN-SOURCE CONTRIBUTION — HOW TO USE THIS MODEL

3-Step Deployment

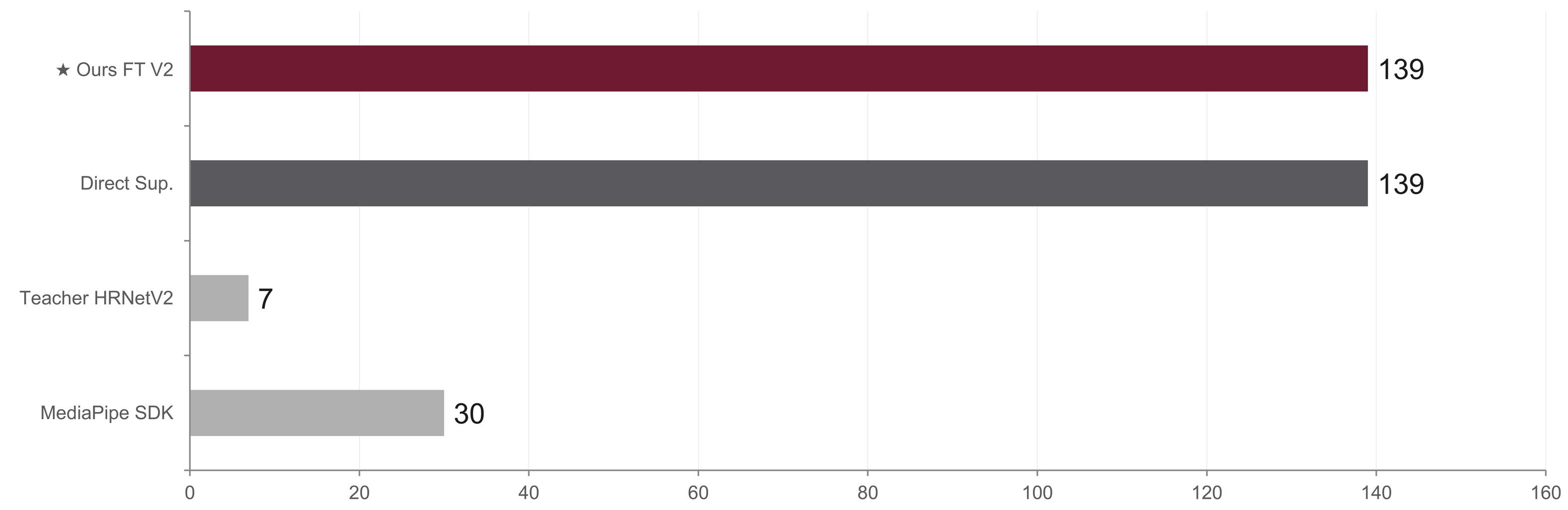
1. **Clone & Install**
   pip install -r requirements.txt
   PyTorch 2.1.0 + MMPose 1.3.2
2. **Load Checkpoint**
   distilled_v2_ft_epoch_150.pth
3. **Run Inference**
   Pass 256×256 RGB tensor
   Get 21×(x,y) keypoints back

Retarget to Any Domain

- **Surgical Tracking** — Fine-tune on OR footage
- **Sign Language** — Domain-specific adaptation
- **Industrial Safety** — Monitor worker hand positions
- **AR / VR Interaction** — Head-mounted device deploy

EFFICIENCY & COMPRESSION

GPU Inference Speed — Models vs. 30 FPS Real-Time Threshold



Batch	Throughput (img/s)
1	126.3
4	473.1
8	891.9
16	905.1

CPU throughput: 25.6 FPS median · Primary limitation for edge-only deployment (GPU recommended)
Batch GPU Throughput — Student (FT V2)

CONCLUSIONS & FUTURE WORK

✓ **KD bridges the representation gap**
MSRAHeatmap codec resolves heatmap→coordinate mismatch without GT labels or architectural changes

✓ **Beats direct supervision**
+3.7 pp AUC · −3.47 px MPJPE · PCK@0.2 = 0.748 · 100% detection rate

✓ **Open, retrainable, deployable**
Any researcher can clone, retrain on new domains, and deploy → no proprietary infrastructure

Future Work
BlazePalm integration (end-to-end pipeline) · ONNX / TFLite / CoreML export · real-world datasets: OneHand10k/ FreiHAND / HO-3D real-world benchmarks

Digital Signal Processing Laboratory

Contact: bea.Juliana.Poquiz@eee.upd.edu.ph